MIL-P-46137A(MR)
13 May 1976
SUPERSEDING
MIL-P-46137(MR)
30 March 1970

MILITARY SPECIFICATION

PLASTIC MOLDING AND EXTRUSION MATERIAL, ACETAL, GLASS FIBER REINFORCED

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of several glass fiber reinforced acetal homopolymer and copolymer compounds suitable for molding and extrusion.

1.2 Classification. The glass fiber reinforced acetal compounds shall be of the following types, grades and classes, as specified (see 6.2, 6.3, and 6.4):

Type I - Higher melting point (homopolymer resin)

Class 20 - 20 percent nominal glass content
Class 30 - 30 percent nominal glass content
Class 40 - 40 percent nominal glass content

Type II - Lower melting point (copolymer resin)

Grade A - Maximum tensile strength material
Class 25 - 25 percent nominal glass content
Grade B - Normal tensile strength material (see 6.9)
Class 20 - 20 percent nominal glass content
Class 30 - 30 percent nominal glass content
Class 40 - 40 percent nominal glass content

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL
PPP-D-723 - Drums, Fiber.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.

FSC - 9330

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STANDARDS

FEDERAL

Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 618 - Conditioning Plastics & Electrical Insulating Materials for Testing
D 638 - Tensile Properties of Plastics
D 648 - Deflection Temperature of Plastics Under Load
D 2133 - Acetal Resin Injection Molding and Extrusion Materials, Specification for
D 2584 - Ignition Loss of Cured Reinforced Resins

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material. The material shall consist of unprocessed glass fiber reinforced acetal homopolymer or copolymer resisn, free from appreciable amounts of other reinforcements or fillers, formulated to comply with the requirements of this specification. There shall be no blending of reinforced and unreinforced material unless approved by the procuring activity (see 6.2). For the purpose of this specification, pelletizing shall not be considered reprocessing. Nominal glass fiber content shall be as specified in 1.2 for class 20, 25, 30 and 40 material.

3.2 Property values. Specimens prepared from the material as specified in 4.3.1 shall conform to the property values specified in table I, when tested by the applicable procedures of 4.3.

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Table I. Property values (see 6.2 and 6.5)

Property	Type I			Type II			
				Grade A	Grade B		
	Class 20	Class 30	Class 40	Class 25	Class 20	Class 30	Class 40
Melting point, °C (°F),							
min.	172(342)	171(340)	171(340)	157(315)	157(315)	157(315)	157(315)
max.	184(363)	183(361)	183(361)	168(335)	168(335)	168(335)	168(335)
Inorganic residue, percent,							
min.	18	28	38	23	18	28	38
max.	22	32	42	27	22	32	42
Inorganic residue, nature of	1/	1/	1/	1/	1/	1/	1/
Tensile strength psi min.	8,000	10,000	10,500	16,000	10,000	10,500	11,000
Tensile modulus, psi x 10^3 min.	800	900	1,000	1,150	850	1,200	1,300
Deflection temperature, (264 psi), min., °C (°F)	145(293)	149(300)	154(310)	152(305)	150(302)	152(305)	154(310)

1/ The inorganic residue shall consist essentially of unfused glass fibers, as determined by visual examination, except that colored compounds may contain non-glass residue equivalent to normal pigmentation. The appearance of this residue shall be uniform between pellets throughout the lot (see table II).

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3.3 Form. The material shall be furnished in the form of pellets or granules (hereinafter referred to as pellets) of the form and size, and containing glass fibers of the diameters and lengths, normally supplied by the manufacturer. In the event that one manufacturer offers glass reinforced acetal products in more than one nominal size or shape of pellet or reinforcement, each such product shall be assigned a different appropriate designation and that designation shall appear on the marking (see 5.2 and 6.4).

3.4 Color. The material shall be of the color and opacity specified. Unless otherwise specified, the color shall be that (a light gray) characteristic of glass fiber reinforced acetal material without added colorant. When close color limits are specified, color matching shall be performed on molded plaques of the thickness specified, in accordance with the method specified by the procuring activity (see 6.2 and 6.6).

3.5 Workmanship. The pellets shall be uniform in color within each container and from container to container. The form and size of both the pellets themselves and of the reinforcement within the pellets shall be reasonably uniform from container to container and shall be uniform within each container within the normal tolerance applicable to the particular method of manufacture. However, any one pellet may contain glass fibers of different diameters and lengths (see notes under tables I and II). The material shall be free of contamination as determined by visual examination.

3.6 Suitability (compatibility) for use with explosives. (Applicable only when specifically designated in the applicable contract, order, drawing, or specification). When so specified, application shall be made to a Government approved laboratory designated by the procuring activity for determination of suitability of the material for use with a particular explosive or explosives (see 6.2 and 6.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all meterial of the same type, grade (if applicable), class, and manufacturer's product designation (if applicable, see 3.3), made by the same process, and submitted for inspection at the same time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, examined, and tested in accordance with the requirements of this specification and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification shall govern. A supplier's certificate of compliance with the requirements specified in 3.1, and the actual glass fiber content shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 50 pounds for examination under 4.2.2.1.1, and in units of shipping containers for examination under 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination shall be approximately 8 ounces.

Table II. Examination of the material for defects in appearance and workmanship

Examine	Defect
Pellets or granules[1/]	Form not as specified (see 3.3) Color not as specified[2/] (see 3.4) Form not uniform (see 3.5) Color not uniform[3/] (see 3.5) Not clean, presence of contamination

1/ If the appearance of the pellets within one sample unit indicates that the unit may contain a mixture of reinforced and unreinforced pellets, this indication shall be verified, if necessary, by picking out a sample of the pellets which are suspected of being unreinforced and of subjecting it to the test for percent inorganic residue. If the appearance of different sample units indicates that they may have been made with different forms of fiberglass reinforcement, that indication shall be verified, if necessary, by ashing separate samples from two units which appear to be different and by comparing the appearances of the two

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inorganic residues. Pellets made with different amounts or forms of reinforcement usually will show characteristic differences in pellet appearance.

2/ This examination shall be for basic color only (natural, white, black, red, blue, etc.), and shall not include exact color matching to a specified standard.

3/ One unit sample selected at random (or a portion of it) shall be used as a standard against which the other units (or portions thereof) shall be compared for uniformity of form and color.

4.2.2.1.2 Examination of packaging. An examination shall be made, in accordance with table III, to determine that preservation, packing and marking comply with specified contract requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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Table III. Examination of packaging

Examine	Defects
Packing	Not level specified; not in accordance with contract requirements. Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability. Inadequate application of components such as: Incomplete closures of case liners; container flaps loose, inadequate strappings, bulged or distorted containers.
Quantity of material	Less than specified or indicated quantity
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete, of improper size, location sequence or method of application, or not in accordance with contract requirements.

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL), expressed as defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

4.2.3 Classification of tests. All tests under this specification shall be classified as lot acceptance tests. Lot acceptance tests shall be made on each lot of material, and in conjunction with the above examination, shall be the basis of acceptance or rejection of the lot.

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4.2.4 Testing. The material shall be tested for the applicable characteristics listed in table I, in accordance with the test methods specified for each lot submitted for inspection. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 50 pounds. The sample unit shall consist of sufficient material to prepare all required specimens. The inspection level shall be S-1 with an acceptance number of 0. The results for each test shall be the averaged results of the specimens, unless only one specimen or determination is specified for testing.

4.3 Test methods.

4.3.1 Preparation of specimens. The tests for melting point (4.3.3) and for percentage and nature of inorganic residue (4.3.4) shall be performed either on the pellets themselves or on pieces cut from molded specimens, except that the verification test for unreinforced pellets, if required (see note 1 of table II), shall be performed only on the pellets. The molded specimens for the above tests, if used, and for all other tests, shall be as described in the applicable test method, and shall be prepared from the material by injection molding in accordance with the supplier's recommendations.

4.3.2 Conditioning. Pellets or granules used directly require no special conditioning, provided that they are reasonably dry. All molded specimens shall be conditioned in accordance with procedure A of ASTM D 618, except that those for the determination of deflection temperature (see 4.3.6) may be annealed. Unless otherwise specified in the test method, all tests shall be conducted at the standard laboratory temperature and in the standard laboratory atmosphere defined in ASTM D 618.

4.3.3 Melting point. One specimen shall be tested in accordance with the melting point procedure specified in ASTM D 2133.

4.3.4 Inorganic residue. Three specimens shall be tested in accordance with ASTM D 2584, with the following exceptions: (a) The percent, by weight, of the non-combustible residue shall be reported, and (b) The appearance of the residue shall be observed for conformance to the inorganic residue, nature requirement of table I.

4.3.5 Tensile strength and modulus of elasticity in tension. Three specimens shall be tested in accordance with ASTM D 638, using type I specimens 1/8 inch thick. The tensile strength and elastic modulus, as defined in that method, shall be reported.

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4.3.6 Deflection temperature under load. Two 1/4-inch thick specimens shall be tested in accordance with ASTM D 648, using a maximum fiber stress of 264 psi. If so desired by the testing agency, or if unannealed specimens do not pass the test, the specimens shall be annealed before testing in accordance with the instructions of the supplier.

5. PACKAGING

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Packing. Packing shall be level A, B, or Commercial (as specified see 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A, in quantities of 200 pounds, maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner propertly heat sealed.

5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 200 pounds, maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size with minimum cube and tare consistent with the protection requirement. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.3 Commercial. The material shall be packed in accordance with Fed. Std. No. 356.

5.1.4 Disposability. (see 6.9).

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5.2 Marking. Unit packages and shipping containers shall be marked in accordance with MIL-STD-129, with the exception that commercial marking in accordance with Fed. Std. No. 356 applies for commercial packaging only. The marking shall include the manufacturer's product designation where required (see 3.3 and 6.4).

6. NOTES

6.1 Intended use. Materials conforming to this specification are used primarily for making injection moldings. Such moldings may be expected to have properties similar to those made from unreinforced acetal resins (see L-P-392 and MIL-HDBK-700), with the added advantage of lower mold shrinkage, better dimensional stability, and higher creep strength at elevated temperatures, but with the disadvantages of lower elongation and somewhat greater processing difficulty. The degree of difference in these properties is a function of glass content.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, grade, and class of material, if required (see 1.2 and 3.2).
(c) Blending of reinforced and unreinforced material, if allowable (see 3.1)
(d) Color and opacity, if required, with method of determination and differences in property requirements, if required (see 3.4, 6.4 and 6.5).
(e) Suitability for use with explosives, if required (see 3.6 and 6.6).
(f) Level of packing required (see 5.1).

6.3 Form. Materials conforming to this specification are available in various pellet lengths containing various lengths and orientations of fiberglass reinforcement, certain of which may be preferable for specific applications. Information concerning these variations may be obtained from the suppliers.

6.4 Colored compounds. The property values specified herein are based on uncolored compounds. Compounds having certain colors can be produced by means of types and concentrations of colorants which do not affect the properties sufficiently to place them outside the specified ranges. However, the use of improper colorants for these colors, or the specifying of colors which can be produced only be means of injurious colorants, could cause the property values to fall outside the limits specified herein. Procurement documents covering colored compounds either should specify only colors which can be produced within the limits of this specification, or should authorize suitable deviations from certain of the property requirements.

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6.5 Color. When a particular color is required, a standard sample, accompanied by borderline standards or by a written statement defining the permissible tolerances and methods of determination, may be furnished by the procuring activity or furnished by the supplier and approved by the procuring activity. As an alternate method, reference may be made to a generally-recognized color standard, such as Fed. Std. No. 595.

6.6 Suitability for use with explosives. Information concerning the suitability of many plastics for use with various explosives and chemicals under various conditions is on file at Picatinny Arsenal. Procuring activities desiring information on this subject should first contact Picatinny Arsenal, Dover, New Jersey 07801 to determine whether the information is already available.

6.7 Design information. Items made of glass fiber reinforced acetal homopolymer and copolymer resins conforming to this specification usually will be found to have specific gravities within the following ranges:

Class	Specific gravity
20	1.51 to 1.59
25	1.58 to 1.65
30	1.60 to 1.67
40	1.68 to 1.75

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6.8 Cross index. The classification of material covered by this specification and the corresponding classification of MIL-P-46137(MR) are as follows:

MIL-P-46137A(MR)	MIL-P-46137(MR)
Type I	Type I
Class 20	Class 20
Class 30	Class 30
Class 40	Class 40
Type II	Type II
Grade A	None
Class 25	None
Grade B	Grade not specified
Class 20	Class 20
Class 30	Class 30
Class 40	Class 40

6.9 Disposability of preservation and packing materials. Environmental pollution preventive measures are contained in the preservation and packing material specifications indicated in section 2. Refer to these specifications for recommended disposability methods.

Custodian:
Army - MR

Review activities:
Army - PA, ME, GL
DSA - GS

User activities:
Army - MI, AT

Preparing activity:
Army - MR

Project No. 9330-A510

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: MIL-P-46137A(MR), Plastic Molding and Extrusion Material, Acetal, Glass Fiber Reinforced

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

SUBMITTED BY (*Printed or typed name and activity*)	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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